Document classification - Simone Pellegrini

```
In [1]: import pandas as pd
        import matplotlib.pyplot as plt
        import numpy as np

        import nltk
        from nltk.corpus import words
        from nltk.tokenize import word_tokenize
        from nltk.stem.wordnet import WordNetLemmatizer
        from nltk.corpus import stopwords as sw
        from sklearn.feature_extraction.text import TfidfVectorizer
        import warnings
        warnings.simplefilter("ignore")
        import os
        os.environ["PYTHONWARNINGS"] = "ignore"
        pd.set_option('display.max_colwidth', None)

        np.random.seed(42)

        label_names = {
            0: "International news",
            1: "Business",
            2: "Technology",
            3: "Entertainment",
            4: "Sports",
            5: "General news",
            6: "Health"
        }
```

Data Exploration

```
In [2]: df = pd.read_csv("development.csv")
        df
```

Out[2]:

	Id	source	title	article	page_rank	tim
0	0	AllAfrica.com	OPEC Boosts Nigeria&#39;s Oil Revenue By .82m Bpd	THE Organisation of Petroleum Exporting Countries (OPEC) is hiking its official output by one million barrels per day effective November with Nigeria getting 82,000 barrels per day or 8.2 per cent of the new quota.	5	20 2
1	1	Xinhua	Yearender: Mideast peace roadmap reaches dead-end in 2004	Looking back at the major events that took place in the Middle East in 2004, the death of veteran Palestinian leader Yasser Arafat is undoubtedly a world-shaker.	5	20 17 1
2	2	Yahoo	Battleground Dispatches for Oct. 5 \\n (CQPolitics.com)\\n	CQPolitics.com - Here are today's Battleground Dispatches Ã¢Â▒Â▒ a roundup of what is going on right now in the year's hottest races, based on reports from local and national media.	5	20 1
3	3	BBC	Air best to resuscitate newborns	Air rather than oxygen should be used to resuscitate newborn babies and may even save lives, research suggests.	5	00 0
4	4	Yahoo	High tech German train crash kills at least one \\n (Reuters)\\n	Reuters - A high tech German train \\ncarrying 29 people smashed into a maintenance vehicle on a \\ntrial run in northern Germany on Friday, killing at least one \\nand trapping 10 more in the wreckage, a local official said. <br clear="all"/>	5	20 1
...	...	...	...	...	...	
79992	79992	Yahoo	Italy's embattled Prodi faces vote of confidence \\n (AFP)\\n	AFP - Embattled Italian Prime Minister Romano Prodi prepared on Wednesday to face the first of two votes of confidence as his centre-left government teetered on the brink of collapse. <br clear="all"/>	5	20 1
79993	79993	All-Baseball.com	Ding Dong, the Deal is Dead	As yesterday began, there was widespread anticipation that the three teams involved in the Johnson trade would send a signed agreement to Commissioner Bud Seligâ▒▒s office so that he could approve the deal.	5	00 0
79994	79994	Yahoo	Two bombs discovered in Sardinia after Berlusconi-Blair meet (AFP)	AFP - Police discovered two bombs near the Sardinian villa of Italian Prime Minister Silvio Berlusconi, just hours after he met his British counterpart Tony Blair on the island, the Ansa news agency said.	5	00 0
79995	79995	Voice	Red Cross Report Alleges US Detainee Abuse at Guantanamo	A report by the International Committee of the Red Cross, quoted Tuesday by the New York Times, accuses the US military of using interrogation tactics that amount to torture against prisoners	5	00 0
79996	79996	Yahoo	UN: Myanmar allows early visit by envoy \\n (AP)\\n	AP - Myanmar's junta has agreed to allow a U.N. envoy to return to the country earlier than expected next month to help promote national reconciliation following the military's violent crackdown on peaceful protesters, the U.N. said Tuesday. <br clear="all"/>	5	20 1

79997 rows × 7 columns

```
In [3]: df = df.drop(columns=["Id"])
```

```
In [4]: df.describe()
```

Out[4]:

	page_rank	label
count	79997.000000	79997.000000
mean	4.892958	2.262760
std	0.397603	1.970502
min	2.000000	0.000000
25%	5.000000	0.000000
50%	5.000000	2.000000
75%	5.000000	4.000000
max	5.000000	6.000000

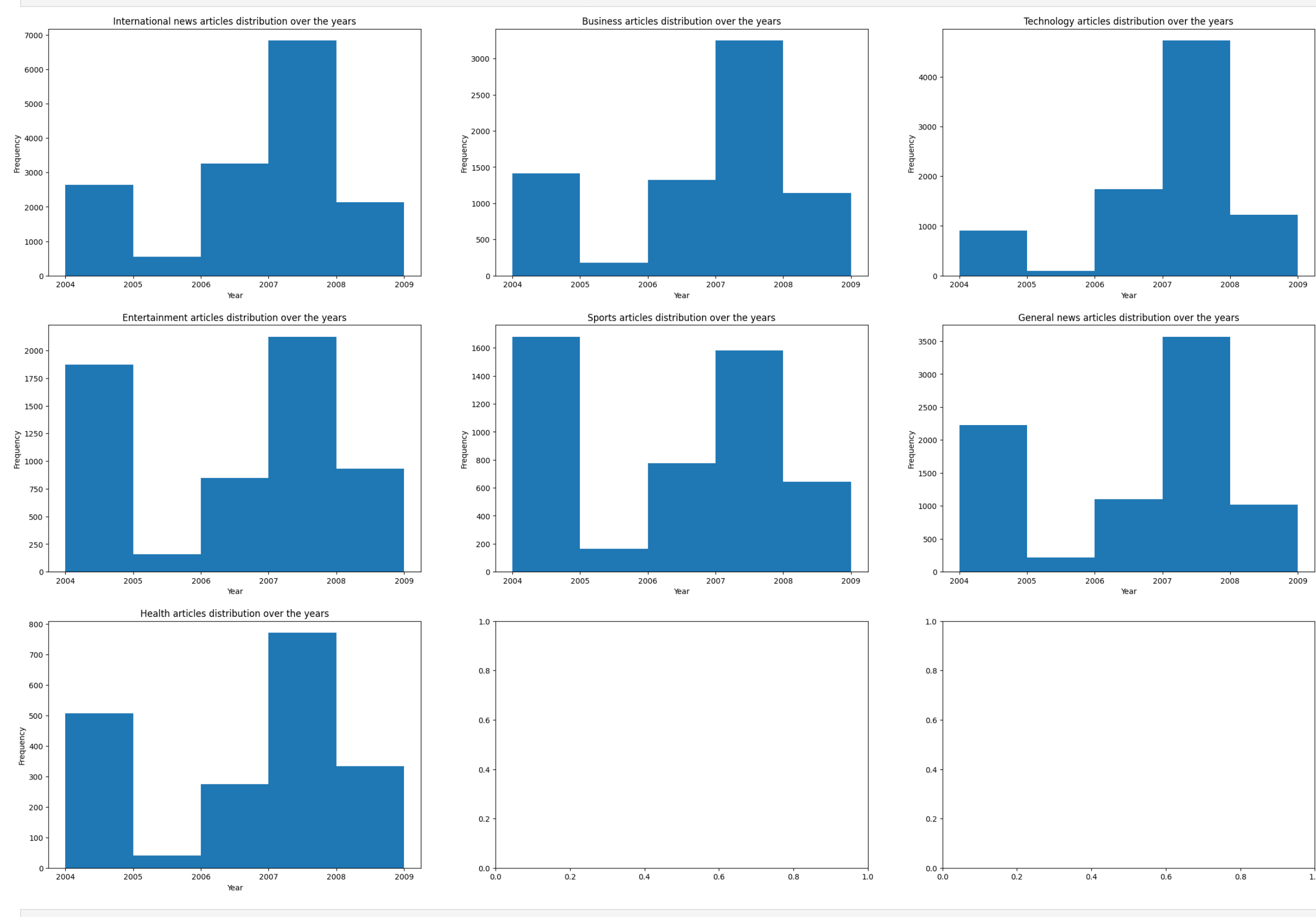
```
In [5]: df = df.dropna()
```

Data exploration

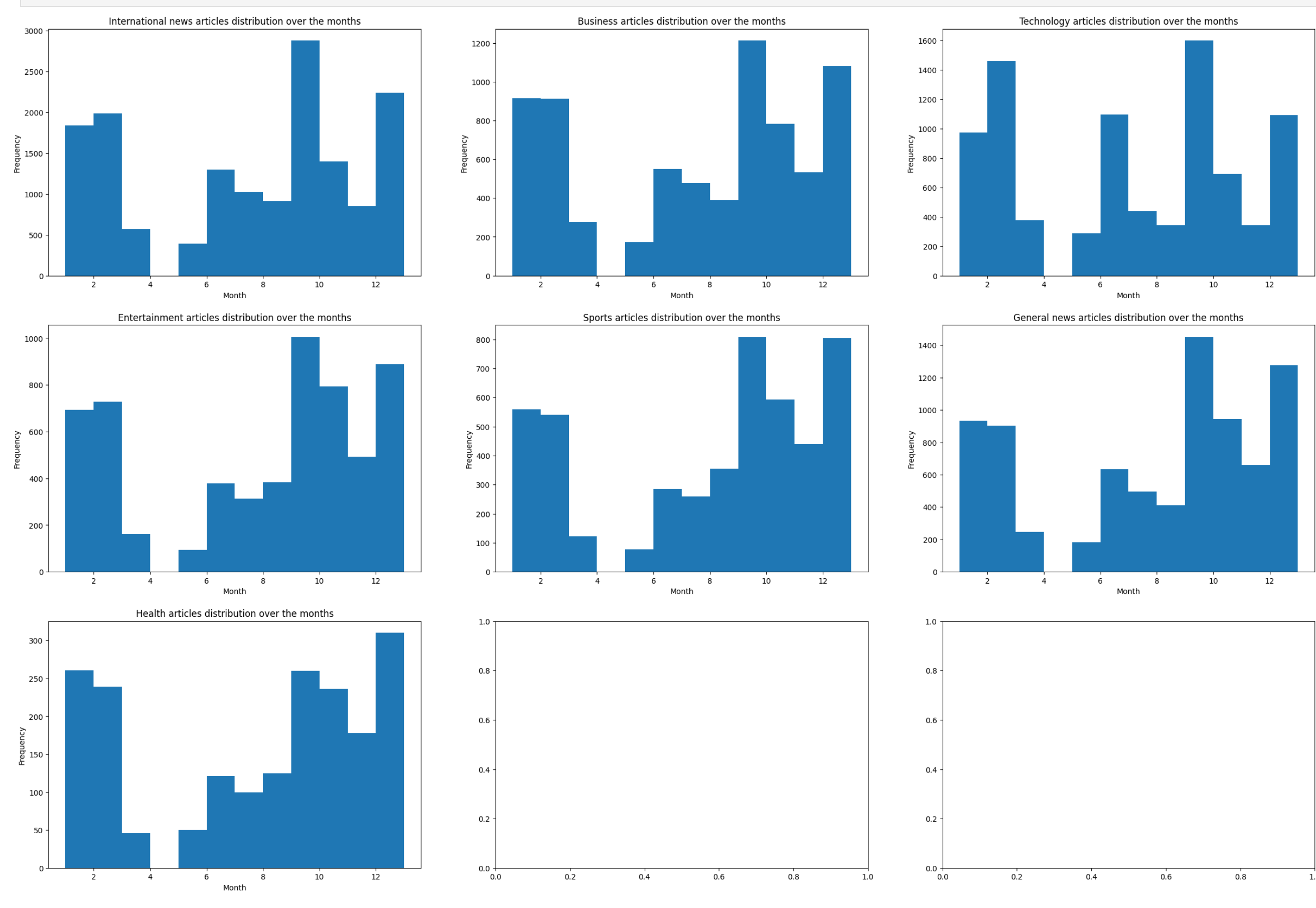
```
In [6]: df["label"].value_counts()
```

```
Out[6]: label
        0    23542
        5    13053
        2    11160
        1    10587
        3     9977
        4     8574
        6     3102
        Name: count, dtype: int64
```

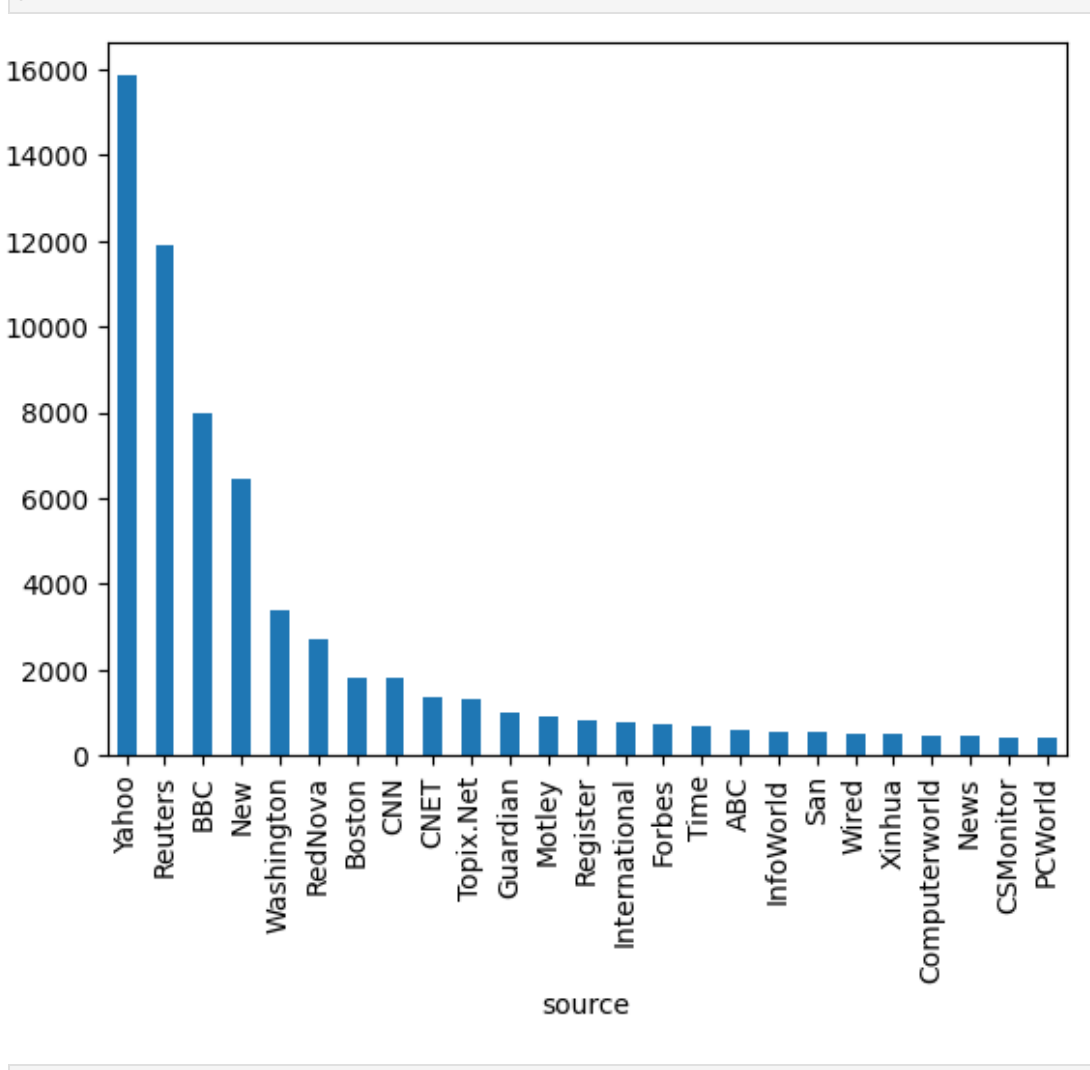
```
In [7]: fig, ax = plt.subplots(3, 3, figsize=(30, 20))
        ax = ax.flatten()

        df_valid = df.copy()
        df_valid["timestamp_dt"] = pd.to_datetime(df_valid["timestamp"], errors="coerce")
        df_valid = df_valid.dropna(subset=["timestamp_dt"])

        df_valid["year"] = df_valid["timestamp_dt"].dt.year
        df_valid["month"] = df_valid["timestamp_dt"].dt.month

        labels = sorted(df_valid["label"].unique())

        for lbl in labels:
            subset = df_valid[df_valid["label"] == lbl]

            ax[lbl].hist(
                subset["year"],
                bins=range(int(subset["year"].min()), int(subset["year"].max()) + 2)
            )
            ax[lbl].set_xlabel("Year")
            ax[lbl].set_ylabel("Frequency")
            ax[lbl].set_title(label_names[lbl] + " articles distribution over the years")
```



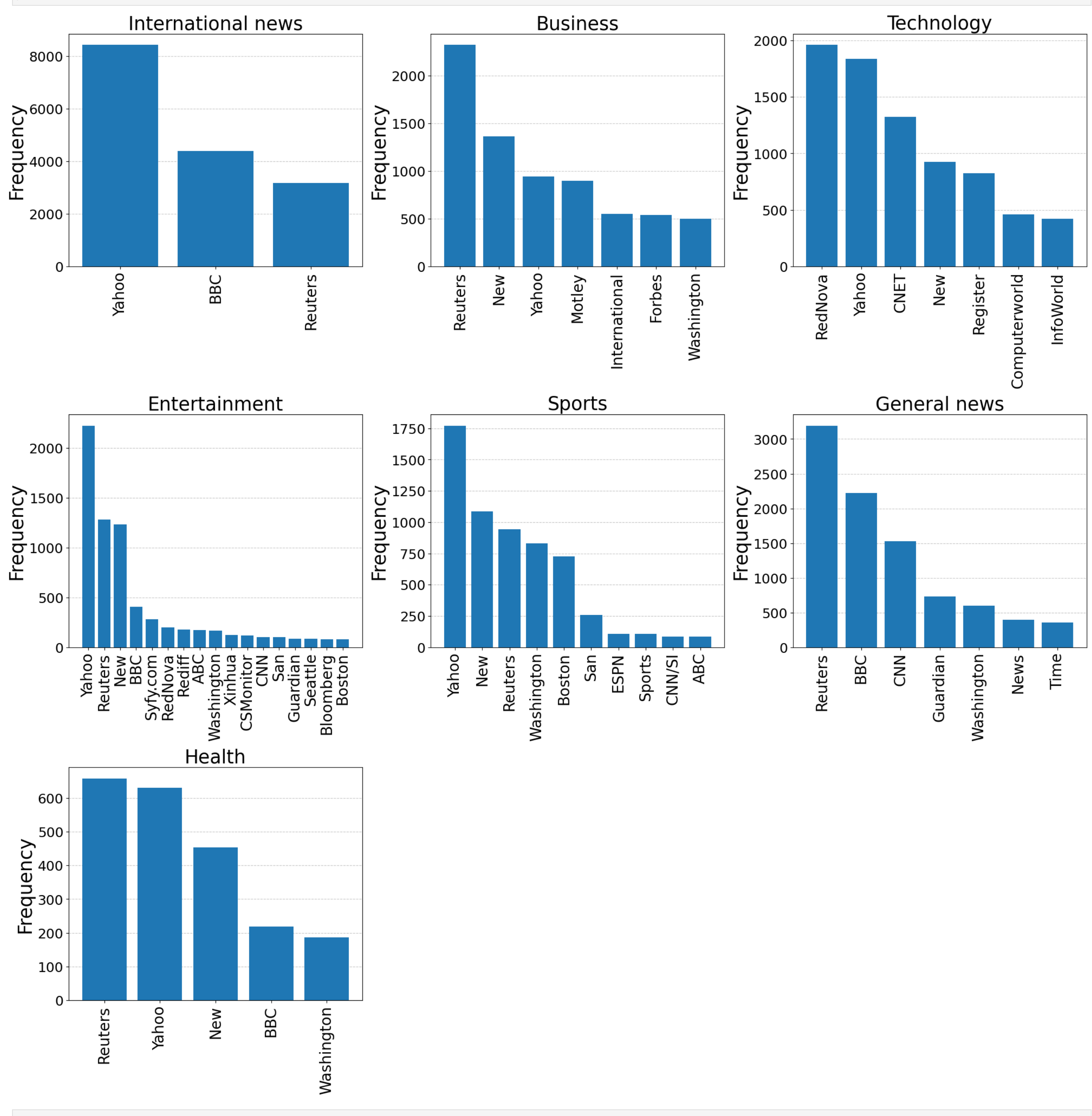
```
In [8]: fig,ax = plt.subplots(3,3,figsize=(30,20))
        ax = ax.flatten()

        for lbl in labels:
            subset = df_valid[df_valid["label"] == lbl]

            ax[lbl].hist(
                subset["month"],
                bins=range(1, 14)
            )
            ax[lbl].set_xlabel("Month")
            ax[lbl].set_ylabel("Frequency")
            ax[lbl].set_title(label_names[lbl]+" articles distribution over the months")
```



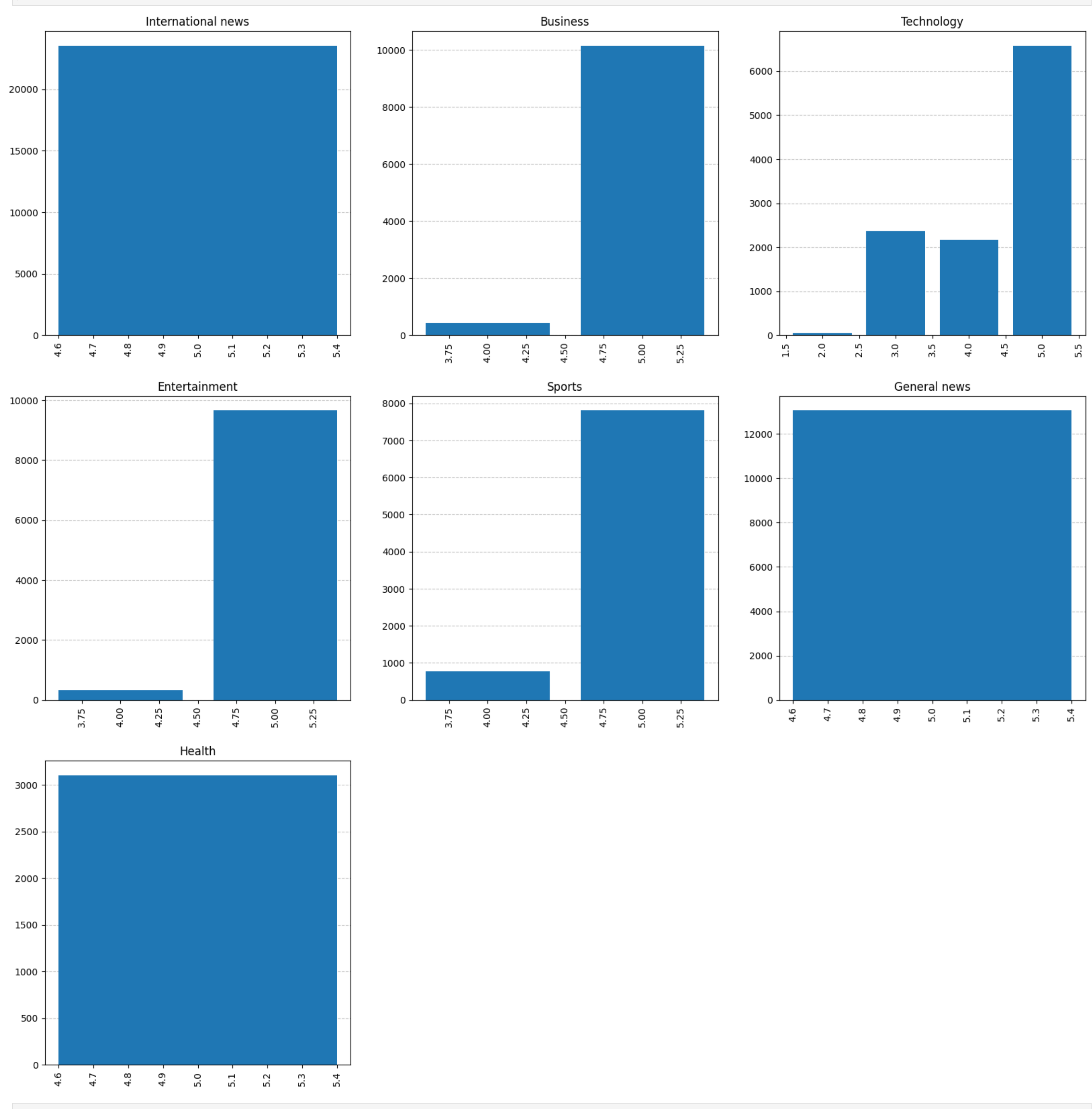
```
In [9]: s = df["source"].value_counts()
        perc = s.cumsum()/df.shape[0]
        s = s[perc<=0.8]
        s.plot(kind='bar')
        plt.show()
```



```
In [10]: def extract_top_k(feature,df,freq,label):
             df = df[df['label']==label]
             s = df[feature].value_counts()
             perc = s.cumsum()/df.shape[0]
             s = s[perc<=freq]
             return s

         fig, ax = plt.subplots(3,3, figsize=(20, 20), constrained_layout=True)
         ax = ax.flatten()

         for p,label in enumerate(range(7)):
             top_k = extract_top_k("source", df, 0.7, label)
             ax[p].set_title(label_names[label], fontsize=25)
             ax[p].bar(top_k.index, top_k.values)
             ax[p].tick_params(axis='x', labelrotation=90,labelsize=20)
             ax[p].tick_params(axis='y', labelsize=18)
             ax[p].grid(axis='y', linestyle='--', alpha=0.7)
             ax[p].set_axisbelow(True)
             ax[p].set_ylabel("Frequency",fontsize=25)

         for i in range(7, len(ax)):
             fig.delaxes(ax[i])

         plt.show()
```



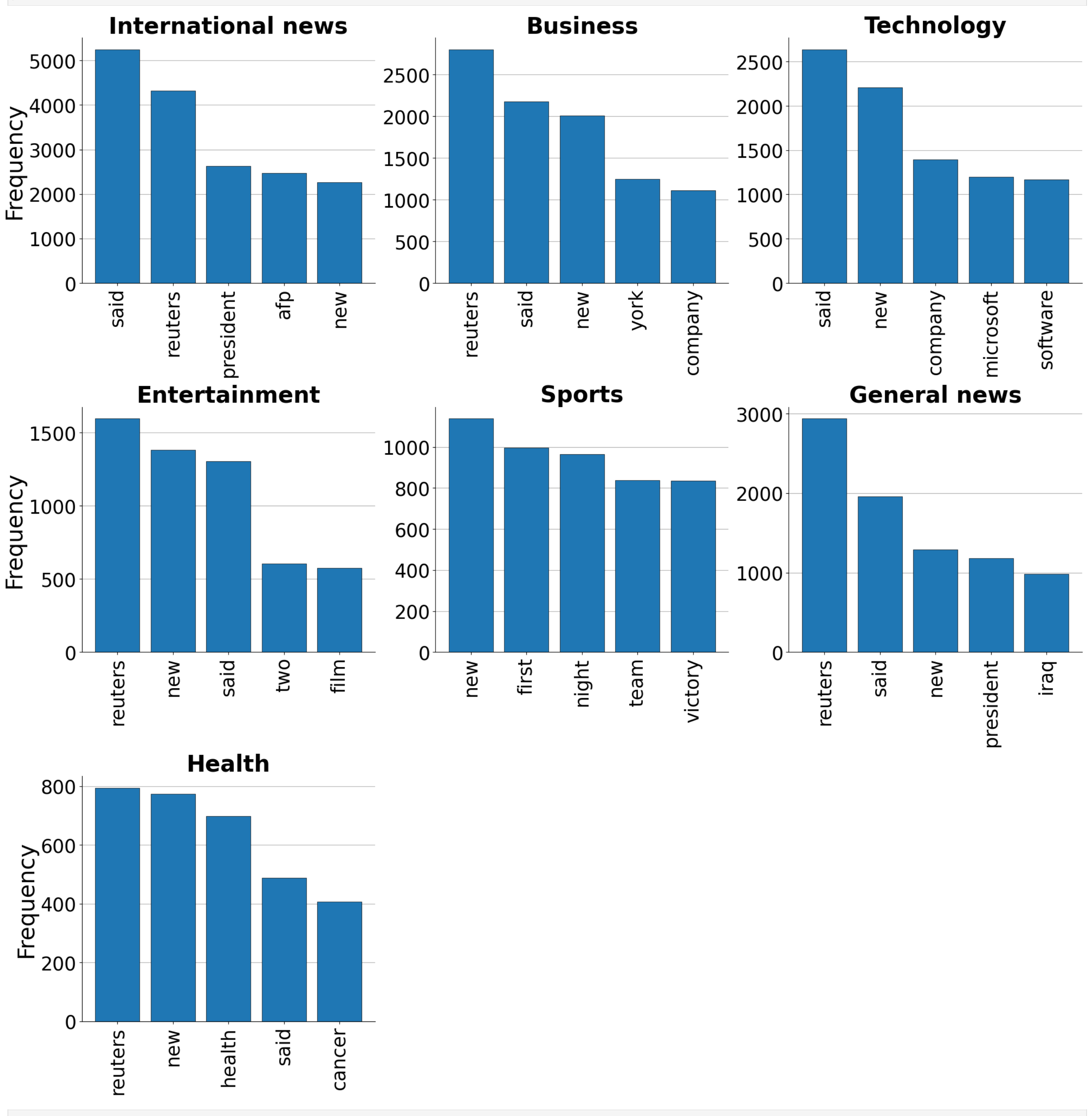
```
In [11]: fig,ax = plt.subplots(3,3,figsize=(20,20))
         ax = ax.flatten()
         for label in range(7):
             s = df.loc[df["label"]==label,"page_rank"].value_counts()
             ax[label].set_title(label_names[label])
             ax[label].bar(s.index,s.values)
             ax[label].tick_params(axis='x', labelrotation=90)
             ax[label].grid(axis='y', linestyle='--', alpha=0.7)
             ax[label].set_axisbelow(True)

         for i in range(7, len(ax)):
             fig.delaxes(ax[i])

         plt.show()
```



```
In [12]: import re

         n_examples = 10
         html_pattern = re.compile(r"<[^>]+>")

         count = 0
         i = 0

         while count < n_examples and 100 + i < len(df):
             article = df["article"][100 + i]

             if isinstance(article, str) and html_pattern.search(article):
                 print("-----------------------")
                 print("LABEL: ", label_names[df["label"][100 + i]])
                 print("TITLE: ", df["title"][100 + i])
                 print("ARTICLE: ", article)
                 print("-----------------------\n")
                 count += 1

             i += 1
```

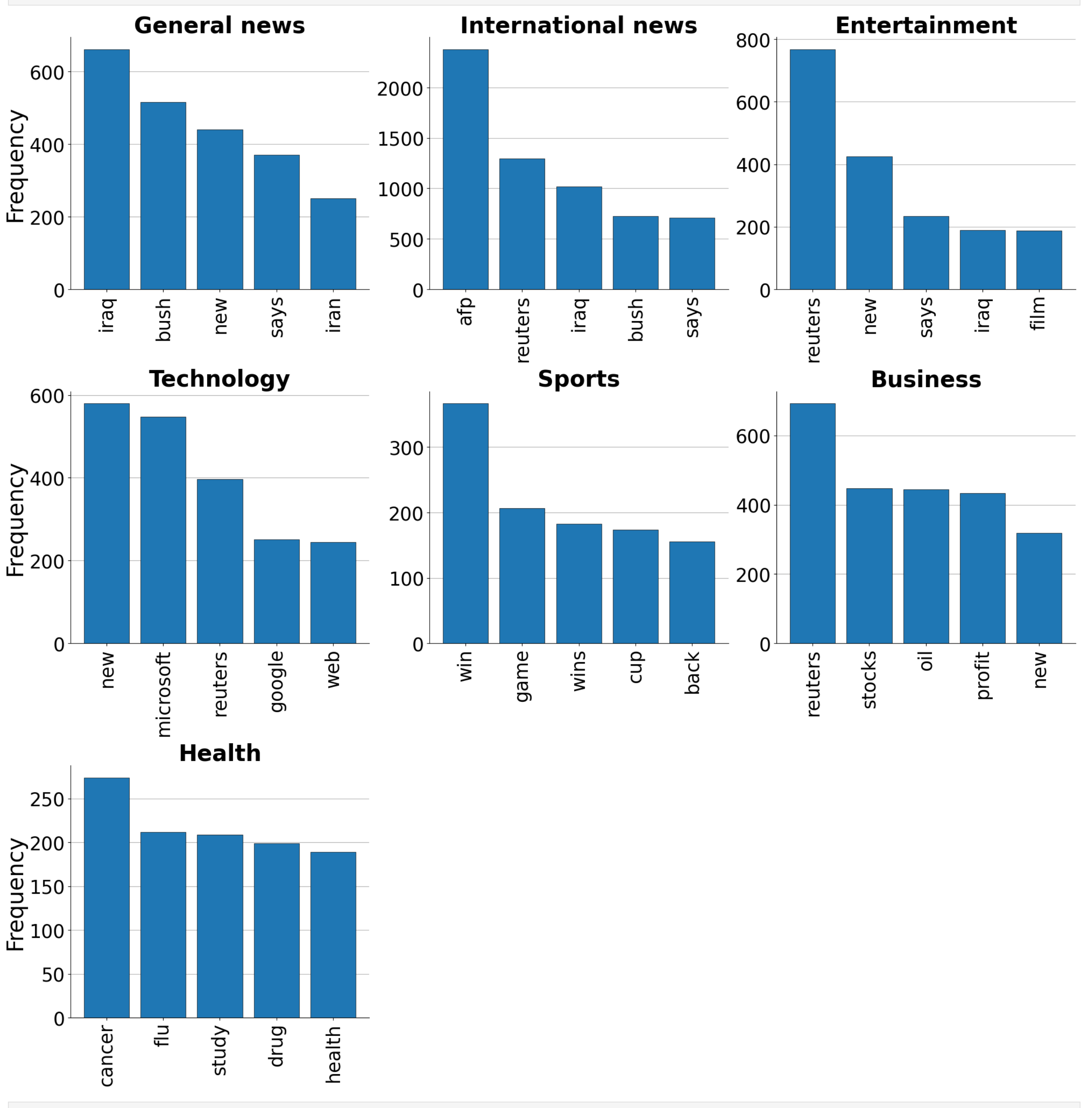
```
-----------------------
LABEL:  International news
TITLE:  Iraq seeks oversight of rape-slaying case \
    (AP)\

ARTICLE:  <p><a href="http://us.rd.yahoo.com/dailynews/rss/world/*http://news.yahoo.com/s/ap/20060704/ap_on_re_mi_ea/iraq"><img src="http://us.news3.yimg.com/us.i2.yimg.co
m/p/ap/20060703/capt.8ec0c90a8a9041e884a7e75dba7940e7.iraq_us_investigation_nccb201.jpg?x=92&amp;y=130&amp;sig=v78JDnM85m.ISPLUiX6ENw--" align="left" height="130" width="9
2" alt="In this photo provided by the Mecklenburg County Sheriff's Office, Steven D. Green is shown in A booking mug shot at the Mecklenburg County jail in Charlotte, N.C.,
Monday, July 3, 2006. Green, a 21-year-old former Army private first class who was recently discharged because of a 'personality disorder,' appeared in a federal magistrat
e's courtroom in Charlotte Monday. The murder and rape charges against him grew out of a military investigation involving up to five soldiers in the March rape and killing
of the woman in Mahmoudiya and three of her relatives, one of them a young girl believed to be about 5 years old. (AP Photo/Mecklenburg County Sheriff's Office)" border="0"
/></a>AP - Iraq&#039;s justice minister demanded Tuesday that the U.N. Security Council ensure a group U.S. troops are punished for allegedly raping and murdering a young I
raqi and executing her family, calling the attack &quot;monstrous and inhuman.&quot;</p><br clear=all>
-----------------------

-----------------------
LABEL:  Technology
TITLE:  Search Engine Forums Spotlight
ARTICLE:  Links to the week's topics from search engine forums across the web: Report: Local Search Advertising Spend Will Double - Content Is King - Do I Need Adwords? - O
rganic Search Vs. Paid Advertisement - SEO Rates - Moving to Pay-For-Performance, and more.<img src="http://feeds.searchenginewatch.com/sew;http://fe~r/sew/~3/http://search
enginewa~d/styles/rss2f~r/sew/~3/http://searchenginewa~d/styles/rss2f~r/sew/~4/525~r/sew/~3/15156943/showP~r/sew/~3/18177873/showP~r/sew/~3/15409354/showP~d/styles/rss2f~r~
r/sew/~3/18177873/showP~d/styles/rss2f~r/sew/~4/19335~r/sew/~4/20281503"/>
-----------------------

-----------------------
LABEL:  Technology
TITLE:  Bank robber exposed by exploding trousers
ARTICLE:  <h4>Smoking pants finger hapless blagger</h4>\
<p>A bank robber who made off with a wad of cash from an Orlando bank last Friday had his collar felt after his trousers exploded as he attempted to make good his escape, t
he <cite>Orlando Sentinel</cite> reports.â▯¦</p>
-----------------------

-----------------------
LABEL:  Technology
TITLE:  Google exceeds expectations in Q4
ARTICLE:  <p>(<a href="http://www.infoworld.com/?source=rss">InfoWorld</a>) - Google beat Wall Street's expectations for its fourth quarter, ended Dec. 31, 2006, as the com
pany continues on a tear, selling search engine advertising much faster than its biggest rivals and than the industry average.</p><p align="right"><a href="http://ad.double
click.net/jump/idg.us.info.rss/news;pos=imu;tile=6;sz=336x280;pkey=business;ord=123456789?" target="_blank" /><img src="http://ad.doubleclick.net/ad/idg.us.info.rss/news;po
s=imu;tile=6;sz=336x280;pkey=business;ord=123456789?" width="336" height="280" border="0" alt="" align="right"/></a></p>\
<p>Google generated revenue of $3.21 billion, an increase of 67 percent compared to the fourth quarter of 2005. Subtracting the commissions it pays to its advertising partn
ers, revenue came in at $2.23 billion, topping the consensus expectation of $2.195 billion from analysts polled by Thomson Financial.</p>\
<p>Net income was $1.03 billion, or $3.29 per share, compared with $372.2 million, or $1.22 per share, in 2005's fourth quarter. On a pro forma basis, which excludes certai
n items, net income was $997 million, or $3.18 per share, topping analysts' expectation of $2.92 per share.</p>
-----------------------

-----------------------
LABEL:  Technology
TITLE:  Last.fm to extend scrobbler tech to video
ARTICLE:  <h4>Building a musical empire, despite DRM</h4>\
<p>Last.fm, the popular music website, is to apply its social recommendation technology to video, as it prepares to do battle with Pandora for the internet radio station ma
rket.â▯¦</p>
-----------------------

-----------------------
LABEL:  General news
TITLE:  Julia Roberts gives birth to twins
ARTICLE:  On <a href='http://www.newsisfree.com/sources/info/2315/'>CNN</a>
-----------------------

-----------------------
LABEL:  International news
TITLE:  Blair to press allies over Afghan forces at EU summit \
    (AFP)\

ARTICLE:  <p><a href="http://us.rd.yahoo.com/dailynews/rss/europe/*http://news.yahoo.com/s/afp/20070307/wl_uk_afp/eusummitbritain"><img src="http://d.yimg.com/us.yimg.com/
p/afp/20070307/capt.sge.dih16.070307135640.photo00.photo.default-512x344.jpg?x=130&y=87&sig=YLsA9a0mNIy4MpfSmhHPUg--" align="left" height="87" width="130" alt="Prime Minist
er Tony Blair leaves 10 Downing Street in London. Blair has urged his NATO allies to bolster troop contributions for Afghanistan, saying he expected the issue to be raised
on the sidelines of this week's EU summit.(AFP/John D McHugh)" border="0" /></a>AFP - Prime Minister Tony Blair urged his NATO allies Wednesday to bolster troop contributio
ns for Afghanistan, saying he expected the issue to be raised on the sidelines of this week's EU summit.</p><br clear="all"/>
-----------------------

-----------------------
LABEL:  International news
TITLE:  Mia Farrow confronts Cambodian police \
    (AP)\

ARTICLE:  <p><a href="http://us.rd.yahoo.com/dailynews/rss/world/*http://news.yahoo.com/s/ap/20080120/ap_on_re_as/cambodia_farrow_darfur"><img src="http://d.yimg.com/us.yim
g.com/p/ap/20080119/capt.ba781475fefb4a7d82f089382e25e3c5.aptopix_cambodia_farrow_darfur_hs102.jpg?x=130&y=84&q=85&sig=Nj2oPVbhwA.3lOG5m1ee8w--" align="left" height="84" wi
dth="130" alt="American actress Mia Farrow takes pictures of  human skulls during her visit at Choeung Ek killing field in Phnom Penh, Cambodia, Saturday, Jan. 19, 2008. Fa
rrow visited a former Khmer Rouge prison Saturday where she plans to hold a ceremony spotlighting the crisis in Sudan, despite a ban imposed by the Cambodian government.
(AP Photo/Heng Sinith)" border="0" /></a>AP - Cambodian police blocked Mia Farrow from holding a genocide memorial ceremony Sunday at a Khmer Rouge prison, at one point for
cefully pushing her group away from a barricade.</p><br clear="all"/>
-----------------------

-----------------------
LABEL:  Entertainment
TITLE:  Is Spring Training a Waste of Time?
ARTICLE:  It may be great for the Arizona and Florida economies, but no one needs six weeks of stretching exercises<img src="http://feeds.feedburner.com/~r/time/entertainme
nt/~4/98580846"/>
-----------------------

-----------------------
LABEL:  Sports
TITLE:  Kotsay helps A's increase lead on Angels \
    (AP)\

ARTICLE:  <p><a href="http://us.rd.yahoo.com/dailynews/rss/sports/*http://news.yahoo.com/s/ap/20060803/ap_on_sp_ba_ga_su/bba_athletics_angels"><img src="http://d.yimg.com/u
s.yimg.com/p/ap/20060802/capt.42d139c0a3654263b0263f1c88794fb6.athletics_angels__baseball_ana101.jpg?x=130&y=87&sig=D9jli7KFd9PfBoUxp5Ddxw--" align="left" height="87" width
="130" alt="Oakland Athletics' Jason Kendall, left, scores against  Los Angeles Angels catcher Jose Molina in the first inning of a baseball game Wednesday, Aug. 2, 2006, i
n Anaheim, Calif. Kendall scored from third base on teammate Frank Thomas' sacrifice fly. (AP Photo/Ric Francis)" border="0" /></a>AP - Taking two of three from the Los Ang
eles Angels, the Oakland Athletics opened a 1 1/2-game lead in an AL West race that has seen all four teams stay in contention going into the final two months.</p><br clear
="all"/>
-----------------------
```

```
In [14]: import math
         from collections import Counter
         from nltk.corpus import stopwords
         from nltk.tokenize import word_tokenize
         from bs4 import BeautifulSoup


         stop_words = set(stopwords.words('english'))
         class_word_counts = {}
         target_col = 'article'

         for label in range(7):
             articles = df[df['label'] == label][target_col]
             all_words = []

             for article in articles:
                 soup = BeautifulSoup(str(article), "html.parser")

                 if soup.body:
                     clean_text = soup.body.get_text(separator=' ')
                 else:
                     clean_text = soup.get_text(separator=' ')

                 clean_text = clean_text.lower()
                 words = word_tokenize(clean_text)
                 words = [w for w in words if w.isalpha() and w not in stop_words and len(w) > 2]
                 all_words.extend(words)

             class_word_counts[label] = Counter(all_words).most_common(5)

         num_classes = len(class_word_counts)
         nrows = 3
         ncols = math.ceil(num_classes / nrows)

         fig, axes = plt.subplots(nrows=nrows, ncols=ncols, figsize=(20, 20), constrained_layout=True)

         axes_flat = axes.flatten()

         for i, (label, words_counts) in enumerate(class_word_counts.items()):
             ax = axes_flat[i]

             if not words_counts:
                 continue

             words, counts = zip(*words_counts)

             ax.bar(words, counts, edgecolor='black', linewidth=0.5)

             title_text = label_names.get(label, label)

             ax.set_title(title_text, fontsize=30, fontweight='bold')

             if i % ncols == 0:
                 ax.set_ylabel("Frequency", fontsize=30)

             ax.tick_params(axis='x', rotation=90, labelsize=25)
             ax.tick_params(axis='y', labelsize=25)

             ax.spines['top'].set_visible(False)
             ax.spines['right'].set_visible(False)

             ax.yaxis.grid(True)
             ax.set_axisbelow(True)

         for j in range(i + 1, len(axes_flat)):
             axes_flat[j].axis('off')

         plt.show()
```



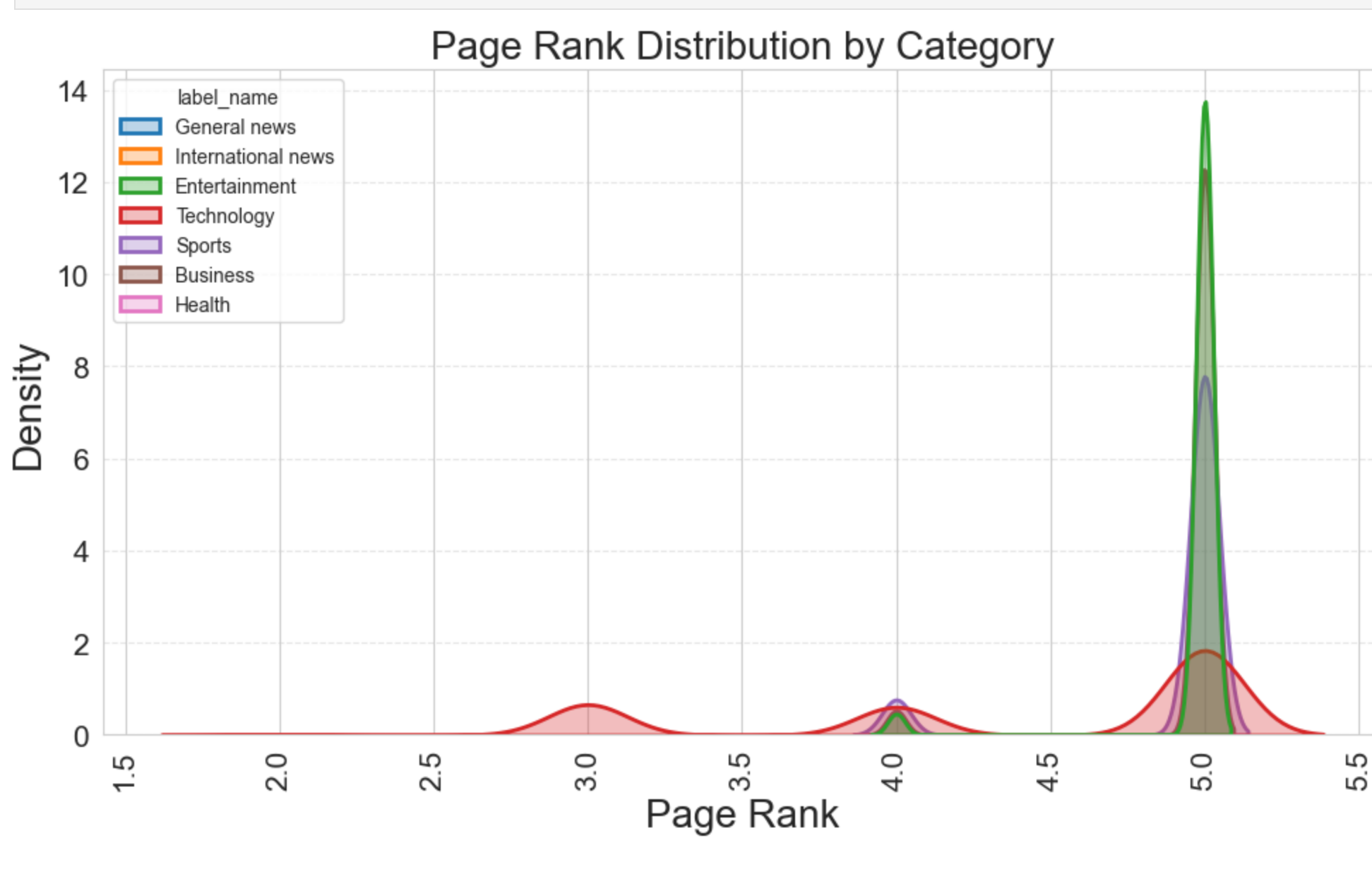
```
In [15]: stop_words = set(stopwords.words('english'))
         class_word_counts = {}
         target_col = 'title'

         for label in df['label'].unique():
             articles = df[df['label'] == label][target_col]
             all_words = []

             for article in articles:
                 article = article.lower()
                 words = word_tokenize(article)
                 words = [w for w in words if w.isalpha() and w not in stop_words and len(w) > 2]
                 all_words.extend(words)

             class_word_counts[label] = Counter(all_words).most_common(5)

         num_classes = len(class_word_counts)
         nrows = 3
         ncols = math.ceil(num_classes / nrows)

         fig, axes = plt.subplots(nrows=nrows, ncols=ncols, figsize=(20, 20), constrained_layout=True)

         axes_flat = axes.flatten()

         for i, (label, words_counts) in enumerate(class_word_counts.items()):
             ax = axes_flat[i]

             if not words_counts:
                 continue

             words, counts = zip(*words_counts)

             ax.bar(words, counts, edgecolor='black', linewidth=0.5)

             title_text = label_names.get(label, label)

             ax.set_title(title_text, fontsize=30, fontweight='bold')

             if i % ncols == 0:
                 ax.set_ylabel("Frequency", fontsize=30)

             ax.tick_params(axis='x', rotation=90, labelsize=25)
             ax.tick_params(axis='y', labelsize=25)

             ax.spines['top'].set_visible(False)
             ax.spines['right'].set_visible(False)

             ax.yaxis.grid(True)
             ax.set_axisbelow(True)

         for j in range(i + 1, len(axes_flat)):
```

```
    axes_flat[j].axis('off')

plt.show()
```



In [100…

```
import seaborn as sns

plt.figure(figsize=(10, 6))

df['label_name'] = df['label'].map(lambda x: label_names[x])

sns.kdeplot(
    data=df,
    x="page_rank",
    hue="label_name",
    fill=True,
    common_norm=False,
    alpha=0.3,
    linewidth=2,
    palette="tab10"
)

plt.title("Page Rank Distribution by Category",fontsize=20)
plt.xlabel("Page Rank",fontsize=20)
plt.ylabel("Density",fontsize=20)
plt.grid(axis='y', linestyle='--', alpha=0.5)
plt.tick_params(axis='x', rotation=90, labelsize=15)
plt.tick_params(axis='y',  labelsize=15)
plt.tight_layout()

plt.show()
```



Preprocessing

In [135…

```
df = df.drop(columns=["timestamp"])
```

In [102…

```
import html

def clean_html(html_content):
    if not isinstance(html_content, str):
        return ""

    soup = BeautifulSoup(html_content, "html.parser")

    for img in soup.find_all("img"):
        src = img.get("src", "")
        alt = img.get("alt", "")
        replacement = f" {alt} {src} ".strip()
        img.replace_with(replacement)

    for a in soup.find_all("a"):
        href = a.get("href", "")
        text = a.get_text(strip=True)
        replacement = f" {text} {href} ".strip()
        a.replace_with(replacement)

    text = soup.get_text(separator=" ")

    text = html.unescape(text)

    text = text.replace("\\", " ")

    text = text.replace("_", " ")

    text = re.sub(r'[()\[\]{}]', ' ', text)

    text = re.sub(r'[^\w\s]', ' ', text)

    return text
```

In [103…

```
class LemmaTokenizer(object):
    def __init__(self):
        self.lemmatizer = WordNetLemmatizer()

    def __call__(self, document):
        lemmas = []
        for t in word_tokenize(document):
            lemma = self.lemmatizer.lemmatize(t)
            lemmas.append(lemma)

        return lemmas

lemmaTokenizer = LemmaTokenizer()
```

In [ ]:

```
def compute_top_k_sources(X, y=None,thresh = 0.8):
    sources = set()
    for i in range(7):
        X_i = X[y==i]
        s = X_i["source"].value_counts()
        perc = s.cumsum()/X_i.shape[0]
        s = s[perc<=thresh]
        sources.update(s.keys())
    return sources

def update_sources(X,top_k_sources):
    X["source"] = X["source"].where(X["source"].isin(top_k_sources), other="other")
```

Model Selection

Balancing the dataset

In [136…

```
idx_zero = df[df["label"] == 0].index
n_drop = int(0.3*len(idx_zero))
drop_idx = np.random.choice(idx_zero, size=n_drop, replace=False)
df_reduced = df.drop(drop_idx)
X = df_reduced.loc[:,["source","title","article","page_rank"]]
y = df_reduced["label"]
```

Parameter tuning of the top k sources

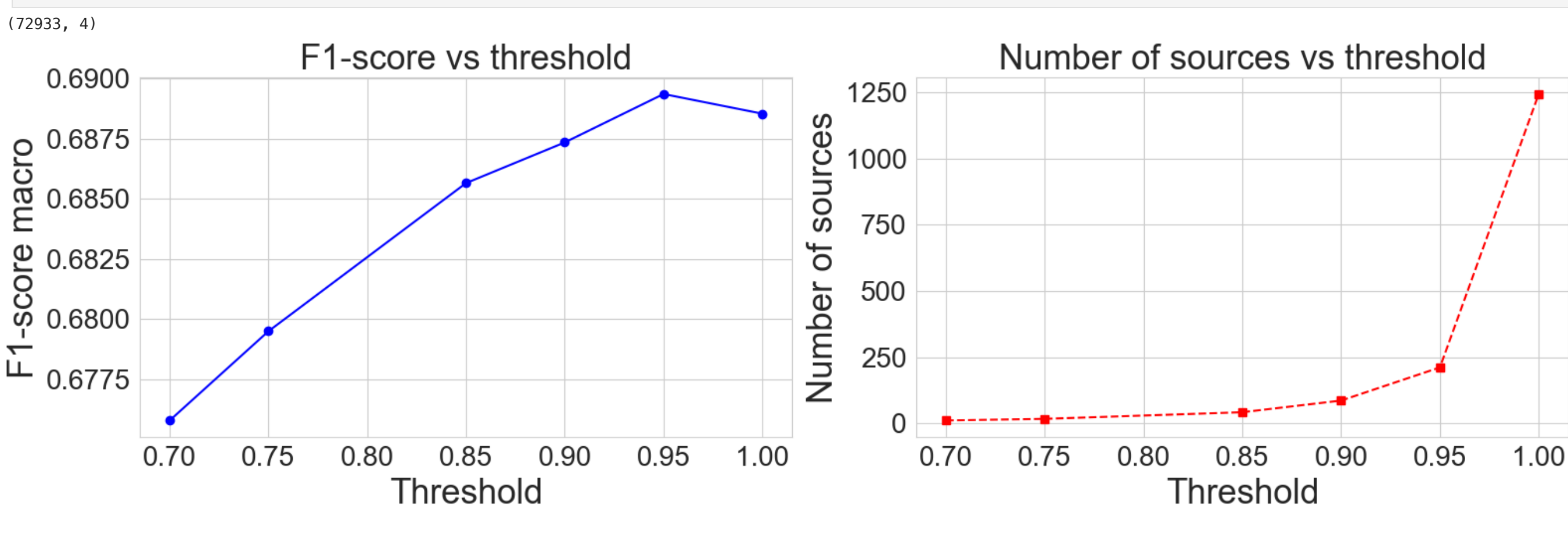
In [106…

```
def extract_top_k_sources(df,freq):
    s = df["source"].value_counts()
    perc = s.cumsum()/df.shape[0]
    s = s[perc<=freq]
    return set(s.index)
```

In [107…

```
from sklearn.svm import LinearSVC
from sklearn.preprocessing import OneHotEncoder
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import StandardScaler
from sklearn.pipeline import Pipeline
from sklearn.model_selection import train_test_split
from sklearn.metrics import f1_score


thresholds = [0.7,0.75,0.85, 0.9,0.95,1]
f1_scores = []
top_k_sizes = []

encoder = OneHotEncoder(handle_unknown='ignore')
scaler = StandardScaler()

preprocessor = ColumnTransformer(transformers = [
    ("title_vectorizer",TfidfVectorizer(),"title"),
    ("article_vectorizer",TfidfVectorizer(),"article"),
    ("encoder",encoder,["source"]),
    ("scaler",scaler,["page_rank"])
])

model = LinearSVC()
pipe = Pipeline(steps=[
    ("preprocessing",preprocessor),
    ("model",model)
])

print(X.shape)
for thresh in thresholds:
    X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2, random_state=42)

    top_k = extract_top_k_sources(X_train, thresh)
    top_k_sizes.append(len(top_k))

    update_sources(X_train, top_k)
    pipe.fit(X_train, y_train)

    update_sources(X_test, top_k)
    f1 = f1_score(pipe.predict(X_test), y_test, average="macro")
    f1_scores.append(f1)

fig, ax = plt.subplots(1, 2, figsize=(16, 5))

ax[0].plot(thresholds, f1_scores, marker='o', linestyle='-', color='b')
ax[0].set_title("F1-score vs threshold",fontsize=25)
ax[0].set_xlabel("Threshold",fontsize=25)
ax[0].set_ylabel("F1-score macro",fontsize=25)
ax[0].tick_params(axis='y', labelsize=20)
ax[0].tick_params(axis='x', labelsize=20)
ax[0].grid(True)

ax[1].plot(thresholds, top_k_sizes, marker='s', linestyle='--', color='r')
ax[1].set_title("Number of sources vs threshold",fontsize=25)
ax[1].set_xlabel("Threshold",fontsize=25)
ax[1].set_ylabel("Number of sources",fontsize=25)
ax[1].tick_params(axis='y', labelsize=20)
ax[1].tick_params(axis='x', labelsize=20)
ax[1].grid(True)

plt.tight_layout()
plt.show()
```

```
(72933, 4)
```



Tuning of the classifier

In [108…

```
from sklearn.pipeline import Pipeline
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.svm import LinearSVC
from sklearn.model_selection import GridSearchCV

preprocessor = ColumnTransformer(
    transformers=[
        ('title_vect', TfidfVectorizer(preprocessor=clean_html,token_pattern=None,  tokenizer=lemmaTokenizer, stop_words="english"), 'title'),
        ('article_vect', TfidfVectorizer(preprocessor=clean_html,token_pattern=None,  tokenizer=lemmaTokenizer, stop_words="english"), 'article'),
        ('source_ohe', OneHotEncoder(handle_unknown='ignore'), ['source']),
        ('scaler', StandardScaler(), ['page_rank'])
    ],
    remainder='drop'
)

pipeline = Pipeline([
    ("preprocessor", preprocessor),
    ("clf", LinearSVC())
])


param_grid = {
    "clf__C": [0.01,0.1,0.3,0.5,0.7,1,1.5,2],
    "clf__class_weight": [None, "balanced"],
    "clf__tol": [1e-4, 1e-8, 1e-12],
    "clf__dual": [True, False]
}


grid = GridSearchCV(
    pipeline,
    param_grid,
    scoring="f1_macro",
    cv=2,
    n_jobs=-1,
    verbose=1
)

grid.fit(X_train, y_train)

print("Best score:", grid.best_score_)
print("Best params:", grid.best_params_)
```

```
Fitting 2 folds for each of 96 candidates, totalling 192 fits
Best score: 0.7046277428653381
Best params: {'clf__C': 0.1, 'clf__class_weight': 'balanced', 'clf__dual': True, 'clf__tol': 0.0001}
```

Tuning of the Tf-Idf parameters

In [109…

```
from sklearn.model_selection import RandomizedSearchCV, train_test_split
from sklearn.svm import LinearSVC
from sklearn.pipeline import Pipeline
from sklearn.compose import ColumnTransformer
from sklearn.preprocessing import OneHotEncoder, StandardScaler
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics import f1_score, make_scorer

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
)
topk = extract_top_k_sources(X_train,0.95)
update_sources(X_train,topk)
update_sources(X_test,topk)

preprocessor = ColumnTransformer(
    transformers=[
        ('title_vect', TfidfVectorizer(preprocessor=clean_html,token_pattern=None,  tokenizer=lemmaTokenizer, stop_words="english"), 'title'),
        ('article_vect', TfidfVectorizer(preprocessor=clean_html,token_pattern=None,  tokenizer=lemmaTokenizer, stop_words="english"), 'article'),
        ('source_ohe', OneHotEncoder(handle_unknown='ignore'), ['source']),
        ('scaler', StandardScaler(), ['page_rank'])
    ],
    remainder='drop'
)

pipe = Pipeline([
    ('preprocessor', preprocessor),
    ('model', LinearSVC(class_weight='balanced',C=0.1,dual=True,random_state=42))
])

param_distributions = {
    'preprocessor__title_vect__max_df': [0.3,0.4,0.5,0.6,0.7,0.8,0.9],
    'preprocessor__title_vect__min_df': [1,2,3,4,5],
    'preprocessor__title_vect__ngram_range': [(1,1), (1,2), (1,3), (1,4),(1,5)],
    'preprocessor__title_vect__max_features': [25000,50000,75000,100000,150000,300000,None],

    'preprocessor__article_vect__max_df':  [0.3,0.4,0.5,0.6,0.7,0.8,0.9],
    'preprocessor__article_vect__min_df': [1,2,3,4,5],
    'preprocessor__article_vect__ngram_range': [(1,1), (1,2), (1,3), (1,4),(1,5)],
    'preprocessor__article_vect__max_features': [25000,50000,75000,100000,150000,300000,None]
}

f1_scorer = make_scorer(f1_score, average='macro')

random_search = RandomizedSearchCV(
    pipe,
    param_distributions=param_distributions,
    n_iter=150,
    scoring=f1_scorer,
    cv=2,
    verbose=1,
    n_jobs=-1,
    random_state=42
)

random_search.fit(X_train, y_train)

print("Best parameters:", random_search.best_params_)
print("Best parameters F1-score (CV):", random_search.best_score_)

y_pred = random_search.predict(X_test)
print("Test F1-score:", f1_score(y_test, y_pred, average='macro'))
```

```
Fitting 2 folds for each of 150 candidates, totalling 300 fits
Best parameters: {'preprocessor__title_vect__ngram_range': (1, 3), 'preprocessor__title_vect__min_df': 1, 'preprocessor__title_vect__max_features': 100000, 'preprocessor__t
itle_vect__max_df': 0.3, 'preprocessor__article_vect__ngram_range': (1, 5), 'preprocessor__article_vect__min_df': 1, 'preprocessor__article_vect__max_features': 100000, 'pr
eprocessor__article_vect__max_df': 0.8}
Best parameters F1-score (CV): 0.7165875806374398
Test F1-score: 0.7312557592105853
```

In [110…

```
random_search.best_params_
```

Out[110…

```
{'preprocessor__title_vect__ngram_range': (1, 3),
 'preprocessor__title_vect__min_df': 1,
 'preprocessor__title_vect__max_features': 100000,
 'preprocessor__title_vect__max_df': 0.3,
 'preprocessor__article_vect__ngram_range': (1, 5),
 'preprocessor__article_vect__min_df': 1,
 'preprocessor__article_vect__max_features': 100000,
 'preprocessor__article_vect__max_df': 0.8}
```

In [111…

```
preprocessor = ColumnTransformer(
    transformers=[
        ('title_vect', TfidfVectorizer(
            preprocessor=clean_html,
            tokenizer=lemmaTokenizer,
            stop_words="english",
            max_df=0.3,
            min_df=1,
            ngram_range=(1, 3),
            max_features=100000
        ), 'title'),

        ('article_vect', TfidfVectorizer(
            preprocessor=clean_html,
            tokenizer=lemmaTokenizer,
            stop_words="english",
            max_df=0.8,
            min_df=1,
            ngram_range=(1, 5),
            max_features=100000
        ), 'article'),

        ('source_ohe', OneHotEncoder(handle_unknown='ignore'), ['source']),
        ("standardizer", StandardScaler(), ['page_rank'])
    ],
    remainder='drop'
)
```

In [137…

```
X_train,X_test,y_train,y_test = train_test_split(X,y,test_size=0.2,random_state=42)
top_k = extract_top_k_sources(X_train,0.95)

update_sources(X_train,top_k)

X_train_transformed = preprocessor.fit_transform(X_train)
print(X_train_transformed.shape)
model = LinearSVC(C=0.1,tol=1e-4,random_state=42,class_weight="balanced",dual=True)
model.fit(X_train_transformed, y_train)
```

```
(58346, 200217)
```

Out[137…

```
LinearSVC
LinearSVC(C=0.1, class_weight='balanced', dual=True, random_state=42)
```

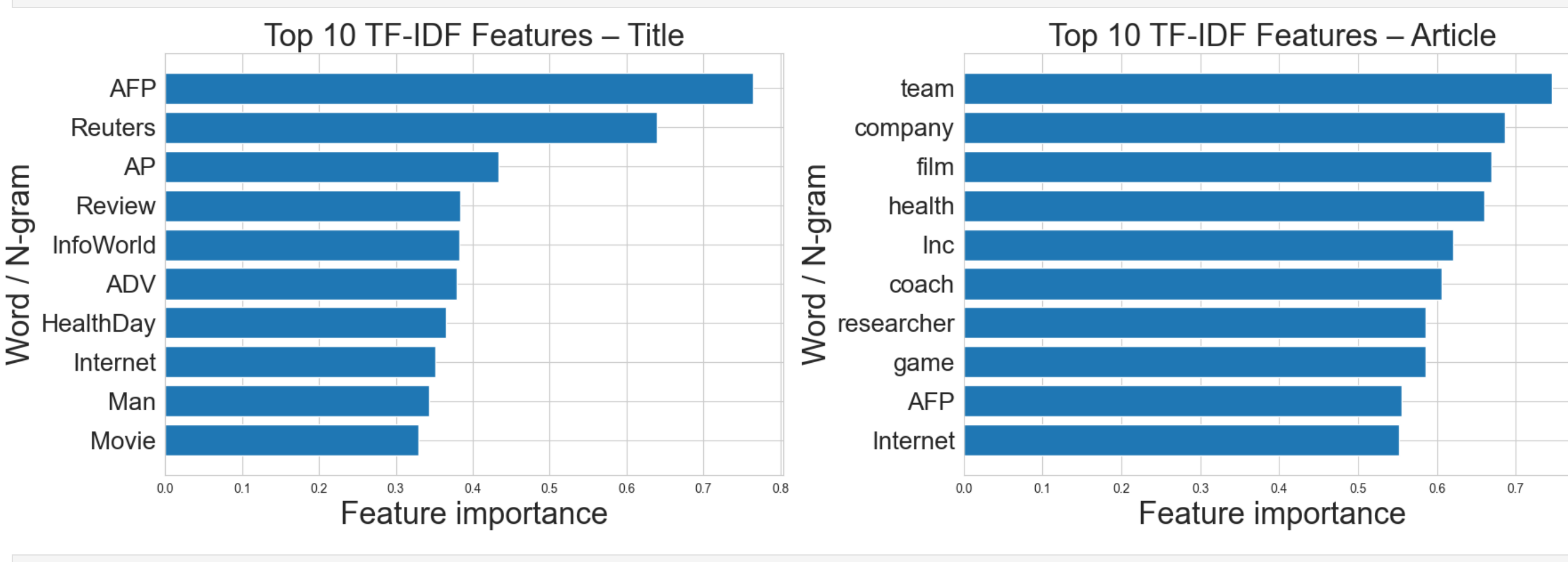
In [138…

```
update_sources(X_test,top_k)
X_test_transformed = preprocessor.transform(X_test)
print(f1_score(model.predict(X_test_transformed),y_test,average="macro"))
```

```
0.7232753217969002
```

In [139…

```
coef = model.coef_


coef = np.mean(np.abs(coef), axis=0)


title_vect = preprocessor.named_transformers_["title_vect"]
article_vect = preprocessor.named_transformers_["article_vect"]

title_features = title_vect.get_feature_names_out()
article_features = article_vect.get_feature_names_out()

n_title = len(title_features)
n_article = len(article_features)


title_coef = coef[:n_title]
article_coef = coef[n_title:n_title + n_article]


def get_top_k(features, weights, k=20):
    idx = np.argsort(weights)[-k:][::-1]
    return features[idx], weights[idx]

top_title_words, top_title_weights = get_top_k(title_features, title_coef, k=10)
top_article_words, top_article_weights = get_top_k(article_features, article_coef, k=10)

fig, axes = plt.subplots(1, 2, figsize=(18, 6))

axes[0].barh(top_title_words[::-1], top_title_weights[::-1])
axes[0].set_title("Top 10 TF-IDF Features – Title",fontsize=25)
axes[0].set_xlabel("Feature importance",fontsize=25)
axes[0].tick_params(axis='y', labelsize=20)
axes[0].set_ylabel("Word / N-gram",fontsize=25)

axes[1].barh(top_article_words[::-1], top_article_weights[::-1])
axes[1].set_title("Top 10 TF-IDF Features – Article",fontsize=25)
axes[1].set_xlabel("Feature importance",fontsize=25)
axes[1].tick_params(axis='y', labelsize=20)
axes[1].set_ylabel("Word / N-gram",fontsize=25)

plt.tight_layout()
plt.show()
```



In [140…

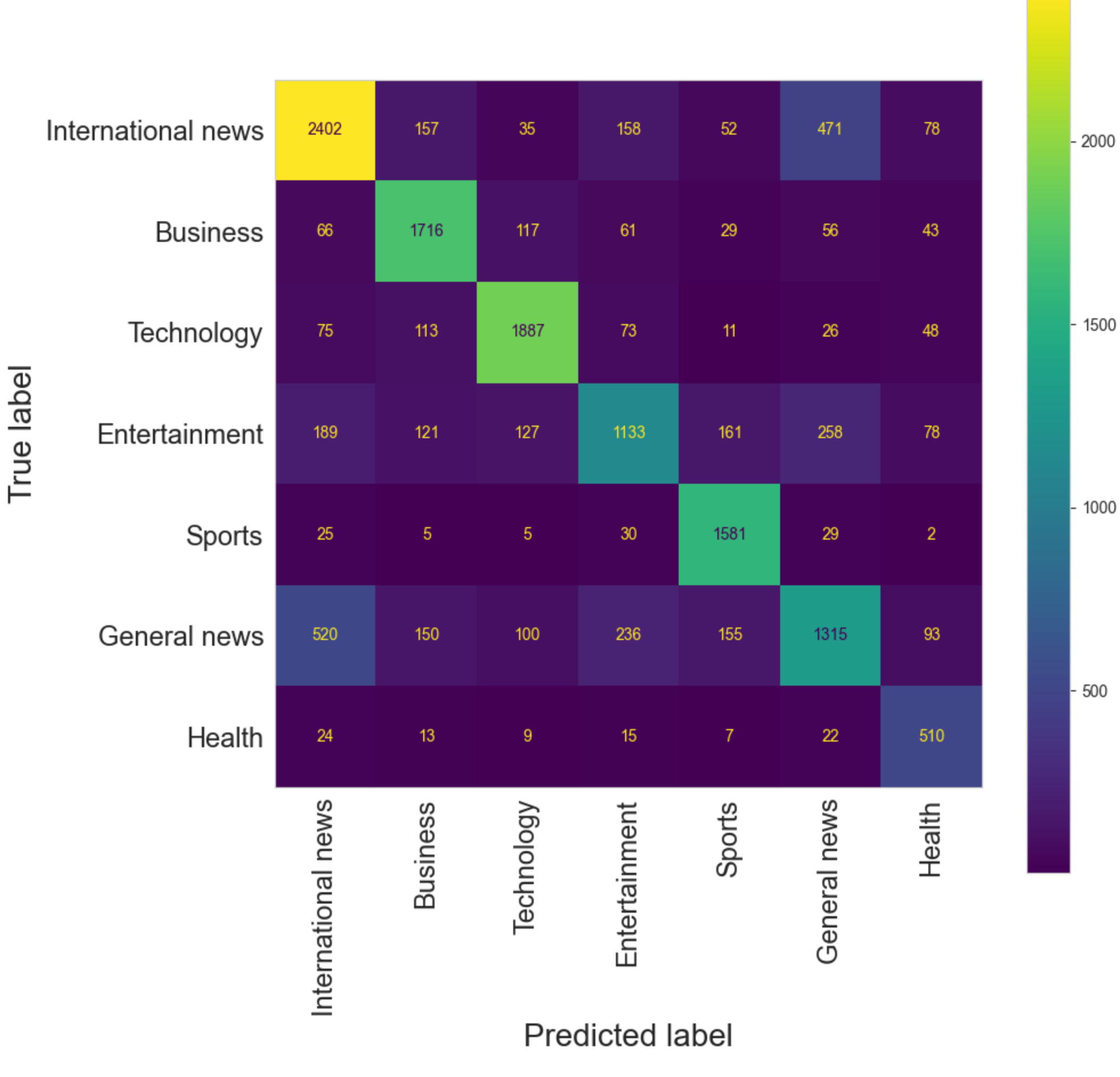
```
update_sources(X_test, top_k)
X_test_transformed = preprocessor.transform(X_test)

y_pred = model.predict(X_test_transformed)

error_analysis = X_test.copy()
error_analysis['True_Label'] = y_test
error_analysis['Predicted_Label'] = y_pred

wrong_predictions = error_analysis[error_analysis['True_Label'] != error_analysis['Predicted_Label']]

print(f"Total errors: {len(wrong_predictions)}")
print("Sample of misclassified rows:")
display(wrong_predictions[0:10])
```

```
Total errors: 4043
Sample of misclassified rows:
```

	source	title	article	page_rank	True_Label	Predicted_Label
79407	BBC	Minimum wage increases to Â£5.35	The minimum wage rate has increased, but many people in the UK do not get the right amount for their work.	5	5	1
41027	Voice	700 Dead, 1,000 Missing from Haiti Flooding	Disaster relief operations are underway in Haiti, where about 700 people were killed and at least 1,000 are missing in flooding spawned by Tropical Storm Jeanne.	5	5	3
22827	Reuters	U.S. Warns of Planned Militant Attacks in Kuwait	KUWAIT (Reuters) - The United States has warned Americans in Kuwait that militants are preparing to launch attacks in the U.S.-allied Gulf Arab state and has urged its citizens to be vigilant.	5	5	0
32402	Reuters	Muslim School Set Ablaze as Dutch Mourn Filmmaker	AMSTERDAM (Reuters) - Arsonists set a Dutch Muslim school ablaze on Tuesday night, the latest in a series of such attacks that came only hours after the cremation of film-maker Theo van Gogh, killed last week by a suspected Islamic militant.	5	3	0
2653	Indian	MI encouraged us to torture, says US soldier	BERLIN, AUGUST 29: A US soldier expected to plead guilty to charges of abusing Iraqi prisoners told a German magazine he deeply regretted his actions but said the abuses were encouraged by Military Intelligence.	5	0	3
30333	New	Identity Badge Worn Under Skin Approved for Use in Health Care	The F.D.A. cleared the way for a company to market implantable microchips that would provide easy access to medical records.	5	2	6
42062	Christian	Free-market Muslims speed Turkey&#39;s march West as leaders declare ...	As a devout Muslim, Celal Hasnalcaci believes women should cover their hair and dress modestly, as the Koran teaches. As	5	0	5
51137	Reuters	TB case prompts new border safety protocol	WASHINGTON (Reuters) - New procedures have been put into place at U.S. borders to stop flagged travelers such as the tuberculosis patient who sneaked across the Canadian border last month, U.S. officials said on Wednesday.	5	0	6
79043	Newsday	Medicare costs may drive away almost everyone	We shouldn&#39;t have been so shocked when the Bush administration sneaked out the news under cover of the Labor Day weekend that the Medicare Part B monthly premium is rising next year by 17.	5	6	5
9367	BBC	Sampanis stripped of bronze	Greek weightlifter Leonidas Sampanis is the first Olympic athlete in Athens to lose his medal for a drugs offence.	5	5	4

In [143…

```
import matplotlib.pyplot as plt
from sklearn.metrics import confusion_matrix, ConfusionMatrixDisplay

y_true = y_test
y_pred = model.predict(X_test_transformed)

target_names = [label_names[i] for i in sorted(label_names.keys())]

cm = confusion_matrix(y_true, y_pred)

fig, ax = plt.subplots(figsize=(10, 10))

disp = ConfusionMatrixDisplay(confusion_matrix=cm, display_labels=target_names)


disp.plot(ax=ax, xticks_rotation='vertical')
ax.xaxis.label.set_size(20)
ax.yaxis.label.set_size(20)
ax.tick_params(axis='both', which='major', labelsize=17)

plt.grid(False)
plt.show()
```



	International news	Business	Technology	Entertainment	Sports	General news	Health
International news	2402	157	35	158	52	471	78
Business	66	1716	117	61	29	56	43
Technology	75	113	1887	73	11	26	48
Entertainment	189	121	127	1133	161	258	78
Sports	25	5	5	30	1581	29	2
General news	520	150	100	236	155	1315	93
Health	24	13	9	15	7	22	510

In [117…

```
y_true = model.predict(X_test_transformed)
wrong_mask = (y_test != y_true)
misclassified = X_test.loc[wrong_mask].copy()

misclassified["y_true"] = y_test[wrong_mask]
misclassified["y_pred"] = y_true[wrong_mask]

misclassified.loc[:, ["title", "article", "y_true", "y_pred"]].head(20)
```

Out[117…

	title	article	y_true	y_pred
79418	Poll blow for Hong Kong democrats	Hong Kong's pro-democracy parties are on course to net fewer seats than expected in legislative elections.	5	0
64051	World response muted as smaller country prepares for 3rd incursion	Smaller but mightier than its neighbors, Rwanda&#39;s greatest clout may come from its nightmare past: a widespread sense around the world that the country was largely abandoned	0	3
27307	E.D.S. President to Become Chief a Little Earlier	Ronald A. Rittenmeyer will replace Michael H. Jordan as chief executive of Electronic Data Systems on September 1, several months earlier than expected.	2	1
28040	Poor start to High Street sales	Boxing Day figures reveal poor sales, but bargain hunters are expected to turn out in force for post-Christmas shopping	0	1
9347	Wine shipments face their day in high court	Terry Speizer wishes he could sell the wines he makes in the cool Edna Valley of California directly to Michigan consumers. In general, alcohol consumed in Michigan must be purchased from	1	3
29778	UK 'has plutonium stockpile'	The UK has built up a stockpile of 100 tonnes of plutonium - enough to make 17,000 nuclear bombs, the Royal Society says.	5	0
51891	Web watch: Google desktop	Google&#39;s Desktop Search application, released as a preview this month for Windows XP and 2000 PCs, searches the hard drive of your PC using the same algorithms that have made its Internet search engine fast, accurate and popular.	3	2
61179	Dick Clark Still Hospitalized	TV personality-producer Dick Clark will spend Christmas, and perhaps New Year&#39;s Eve, in a hospital where he&#39;s recovering from a mild stroke, his spokesman said Tuesday.	5	3
25116	Update 4: Blockbuster to Make Hollywood Hostile Bid	Blockbuster Inc. said Tuesday it will launch a hostile bid for Hollywood Entertainment Corp. next month if the rival&#39;s directors won&#39;t negotiate a deal.	3	1
20593	Roman empires to collide	Chelsea&#39;s 3-1 win against FC Porto was not owner Roman Abramovich&#39;s only reason to smile after Champions League action on Wednesday.	5	4
79729	Ontario aims to end smoking in Legions, clubs	TORONTO - Royal Canadian Legion halls and so-called private clubs will have to butt out under anti-smoking legislation the Ontario government plans to introduce this month, Health Minister George Smitherman said Tuesday.	5	6
31711	New Chief at Pfizer Will Reduce Sales Force	The worldÂ▯s largest drug company said that it would lay off almost 2,400 employees in what may indicate a wider retrenchment of the drug industry.	6	1
13770	Sony Computer to launch worldwide sales of new PS2 console in Nov	TOKYO, Sep 21, 2004 (AFX-UK via COMTEX) -- TOKYO (AFX) - Sony Computer Entertainment Inc, the game unit of Sony Corp, will roll out worldwide a new, smaller version of the PlayStation 2 game console from the start of November, company President Ken	2	1
29116	Elvis announcer dies in car crash	The man famed for saying "Elvis has left the building" dies in a crash after leaving an Elvis convention.	3	0
8770	The Saturday Profile: A U.S. Diplomat With an Extraordinary Global View	Americaâ▯▯s first blind diplomat, Avraham Rabby, spent the last two years surveying Trinidadâ▯▯s political landscape and composing cables to Washington in Braille.	0	1
24425	Fiat and GM to mull car unit fate	Fiat will meet car giant GM on Tuesday in an attempt to reach agreement over the future of the Italian firm's loss-making auto group.	0	1
54638	Dance: Alvin Ailey Company Flexes New Muscle	The company has transcended the trauma of its founderÂ▯s death and has forged into the future with a strong roster of dancers and a steady touring schedule.	0	3
28272	Suspected North Korean refugees enter Canadian embassy in China	A group of suspected North Korean refugees has broken into the Canadian embassy in the Chinese cpaital, Beijing. Our China correspondent, John Taylor, reports the group on Thursday afternoon scaled the walls of the embassy compound.	5	0
33930	Israeli Soldier Killed in Gaza Blast - Sources	An Israeli soldier was killed and four were wounded in a bomb attack carried out by Palestinian militants in the central Gaza Strip on Tuesday, Israeli military sources said.	5	0
32445	Hamm Ruling Stands, but Ire at Judges Rises	The results of the Olympic gymnastics all-around final will not change, leaving Paul Hamm of the United States with the sole gold medal.	0	4

In [119…

```
df = pd.read_csv("evaluation.csv")
X = df.loc[:,["source","title","article","page_rank"]]

X["title"] = X["title"].fillna("")
X["article"] = X["article"].fillna("")

update_sources(X,top_k)
X_transformed = preprocessor.transform(X)
y=model.predict(X_transformed)
```

In [ ]:

```
submission = pd.DataFrame({
    'Id': range(len(y)),
    'Predicted': y
})

submission.to_csv("submission.csv", index=False)

print("File 'submission.csv' created succesfully!")
```

```
File 'submission.csv' created succesfully!
```